

2.14 understand the importance in plants of turgid cells as a means of support

- 2.15 understand the factors that affect the rate of movement of substances into and out of cells to include the effects of surface area to volume ratio, temperature and concentration gradient
- 2.16 describe simple experiments on diffusion and osmosis using living and non-living systems.

e) Nutrition

Students will be assessed on their ability to:

Flowering plants

- 2.17 describe the process of photosynthesis and understand its importance in conversion of light energy to chemical energy
- 2.18 recall the word equation and the balanced chemical symbol equation for photosynthesis
- 2.19 understand how carbon dioxide concentration, light intensity and temperature affect the rate of photosynthesis
- 2.20 explain how the structure of the leaf is adapted for photosynthesis
- 2.21 recall that plants require mineral ions for growth and that magnesium ions are needed for chlorophyll and nitrate ions are needed for amino acids
- 2.22 describe simple controlled experiments to investigate photosynthesis, showing the evolution of oxygen from a water plant, the production of starch and the requirements of light, carbon dioxide and chlorophyll

Humans

- 2.23 understand that a balanced diet should include appropriate proportions of carbohydrate, protein, lipid, vitamins, minerals, water and dietary fibre**
- 2.24 recall sources and describe functions of carbohydrate, protein, lipid (fats and oils), vitamins A, C and D, and the mineral ions calcium and iron, water and dietary fibre as components of the diet
- 2.25 understand that energy requirements vary with activity levels, age and pregnancy**
- 2.26 recognise the structures of the human alimentary canal and describe in outline the functions of the mouth, oesophagus, stomach, small intestine, large intestine, and pancreas
- 2.27 understand the processes of ingestion, digestion, absorption, assimilation and egestion
- 2.28 explain how and why food is moved through the gut by peristalsis
- 2.29 understand the role of digestive enzymes to include the digestion of starch to glucose by amylase and maltase, the digestion of proteins to amino acids by proteases and the digestion of lipids to fatty acids and glycerol by lipases
- 2.30 recall that bile is produced by the liver and stored in the gall bladder, and understand the role of bile in neutralising stomach acid and emulsifying lipids
- 2.31 explain how the structure of a villus helps absorption of the products of digestion in the small intestine
- 2.32 recall how to carry out a simple experiment to determine the energy content in a food sample.**

f) Respiration

Students will be assessed on their ability to:

- 2.33 recall that the process of respiration releases energy in living organisms
- 2.34 describe the differences between aerobic and anaerobic respiration
- 2.35 recall the word equation and the balanced chemical symbol equation for aerobic respiration in living organisms
- 2.36 recall the word equation for anaerobic respiration in plants and in animals
- 2.37 describe simple controlled experiments to demonstrate the evolution of carbon dioxide and heat from respiring seeds or other suitable living organisms.**

g) Gas exchange

Students will be assessed on their ability to:

- 2.38 understand the role of diffusion in gas exchange